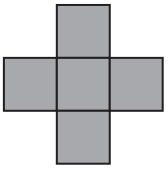
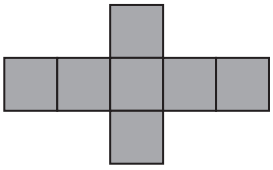
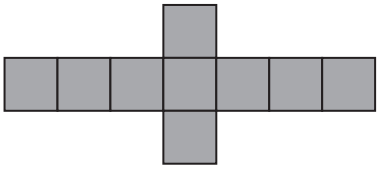
13 Here is a sequence of patterns made with grey squares.



pattern number 1

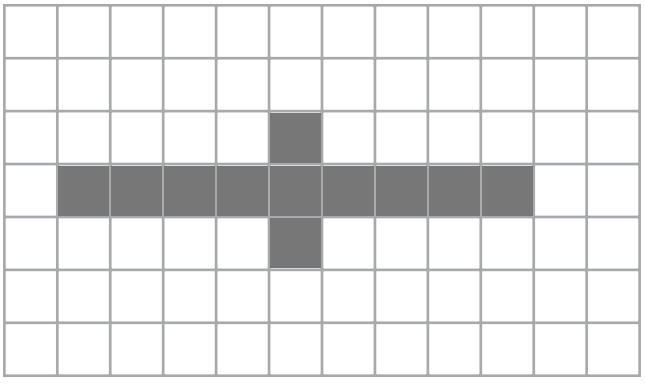


pattern number 2



pattern number 3

(a) On the grid, draw pattern number 4



1 mark
Extend both ends by one square.
9 across + 2 = 11.

(1)

(b) Find the total number of grey squares in pattern number 7

5, 7, 9, 11, 13, 15, 17

Pattern 7: 5 + 6 × 2 = 17

17

(2)

2 marks total
Add 2 for each step; six steps from pattern 1 to pattern 7.

In pattern number 100 there are 203 grey squares.

(c) Find the total number of grey squares in pattern number 99

Pattern 99: 203 – 2 = 201

201

(1)

1 mark
Step back once from pattern 100.

(Total for Question 13 is 4 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS